Strategic Form Games: Some Examples

The Prisoners' Dilemma (G_{PD})

Player 1 \ Player 2	C	D
C	$2, 2$	$-1, 3$
D	$3, -1$	$0, 0$

Battle of the Sexes (G_{BS})

Player 1 \ Player 2	O	W
O	$1, 2$	$0, 0$
W	$0, 0$	$2, 1$

Coordination Failure (G_{CF})

Player 1 \ Player 2	R	S
R	$2, 2$	$0, 1$
S	$1, 1$	$1, 1$

Matching Pennies (G_{MP})

Player 1 \ Player 2	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

Two games illustrating IDSDS and Rationalizability.

Player 1 \ Player 2	L	M	R
U	$0, 2$	$0, 1$	$1, 0$
D	$1, 1$	$1, 2$	$0, 0$

The preceding game has a unique $IDSDS$ profile, (D, M).

Player 1 \ Player 2	L	M	R
U	$0, 0$	$0, 3$	$1, 2$
D	$1, 3$	$1, 0$	$0, 2$

For this game, no profiles are deleted by $IDSDS$. But (D, L) is the unique (point) Rationalizable strategy.

The Cournot Game, G_C. Let $n \geq 2$. Set $S_i = \mathbb{R}_+$ (output) and for each i let $\succsim_i$ be represented by u_i (i's profit) given by

$$u_i(s) = p\Big(\sum_{j}^{n} s_j\Big)s_i - c_i(s_i),$$

where $p(\cdot)$ is an inverse demand and c_i is firm i's cost function. The inverse demand and cost functions satisify the usual conditions we imposed for a monopoly ($n = 1$). Question: For $n \geq 2$, when is $DSE(G_C)$ not empty?